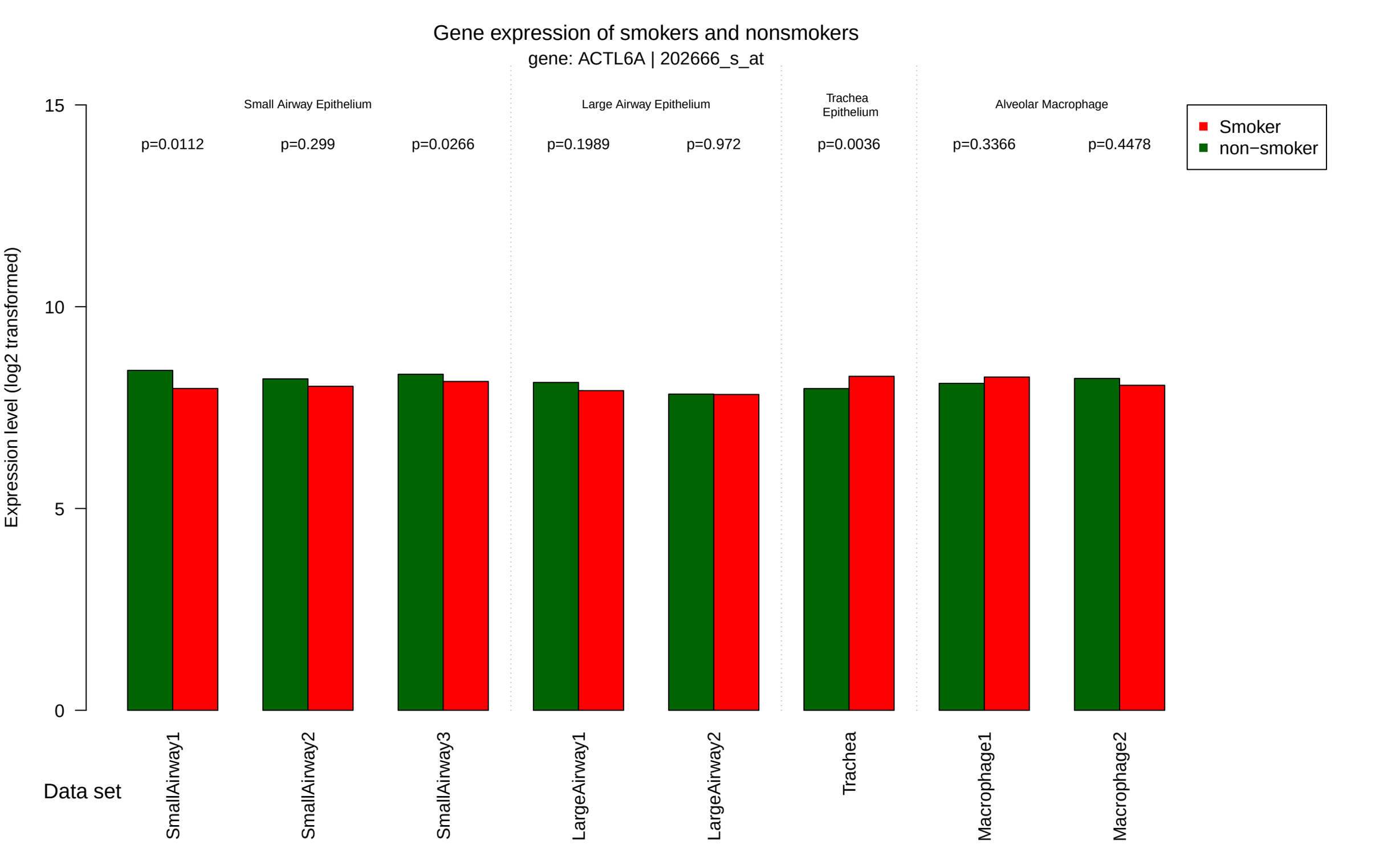
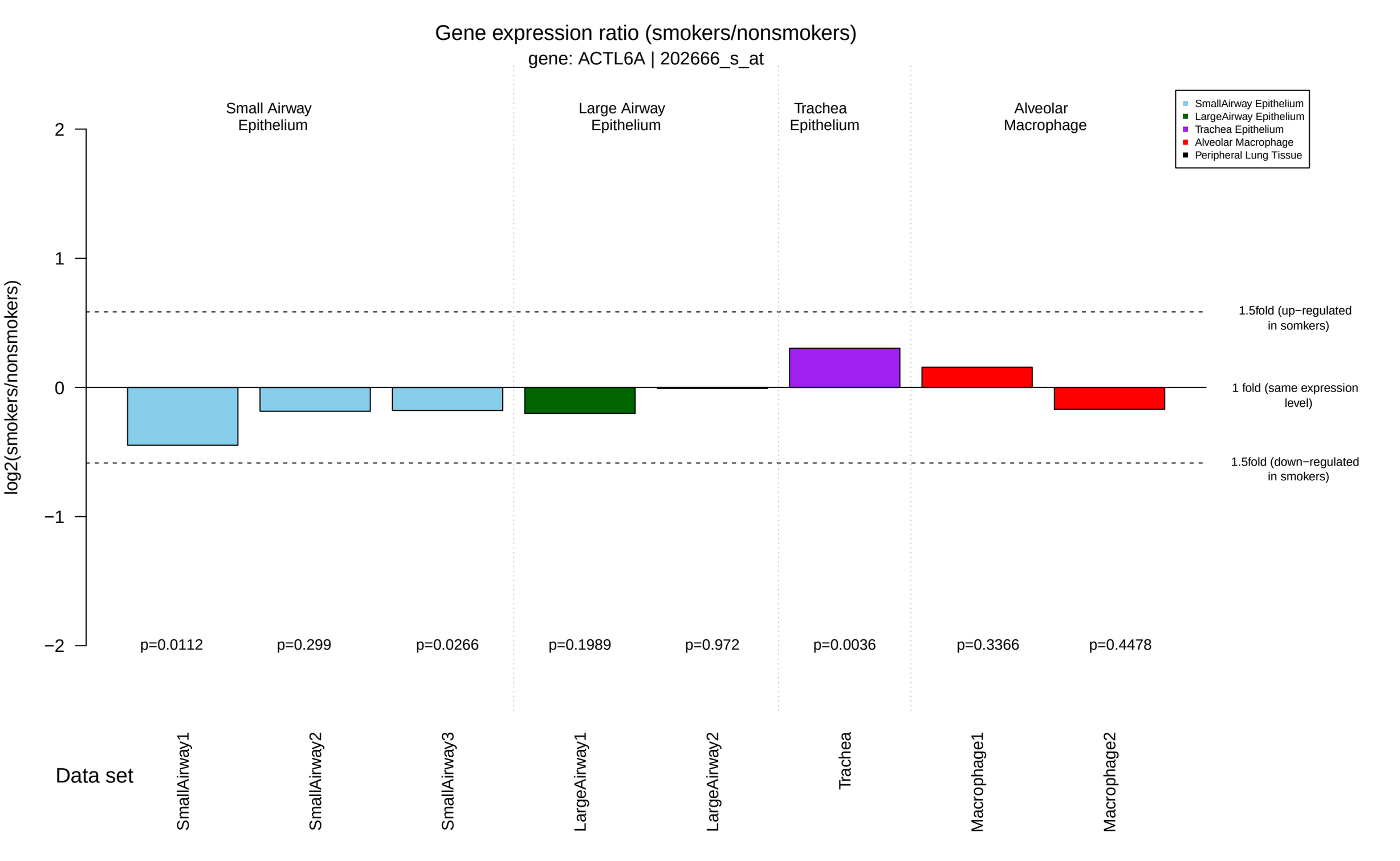
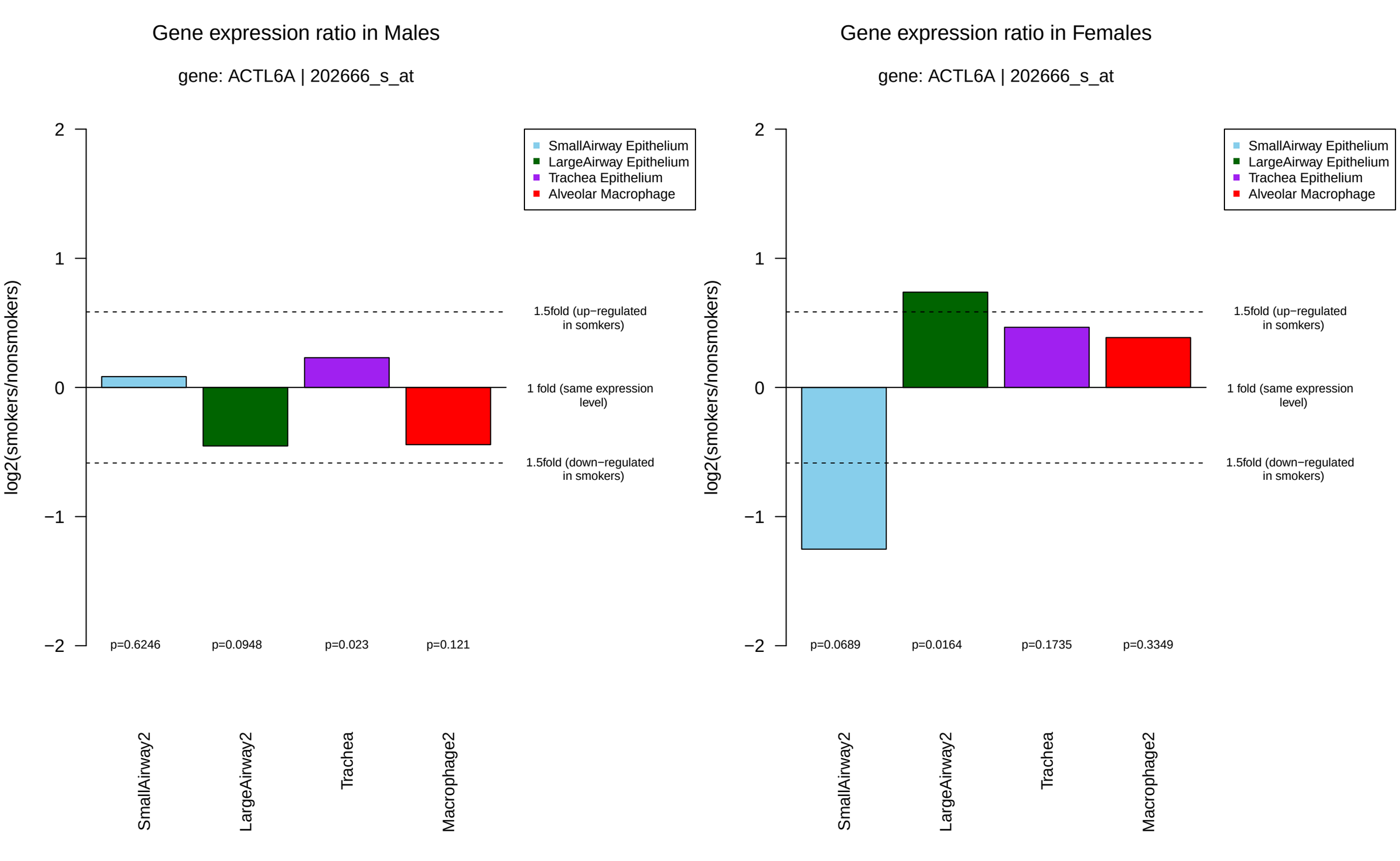
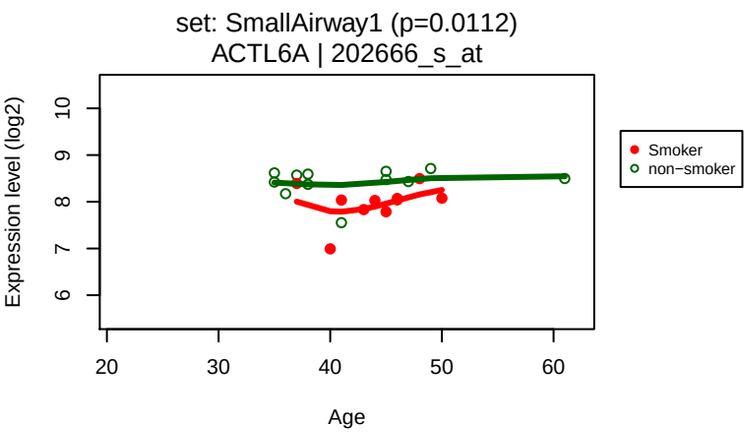
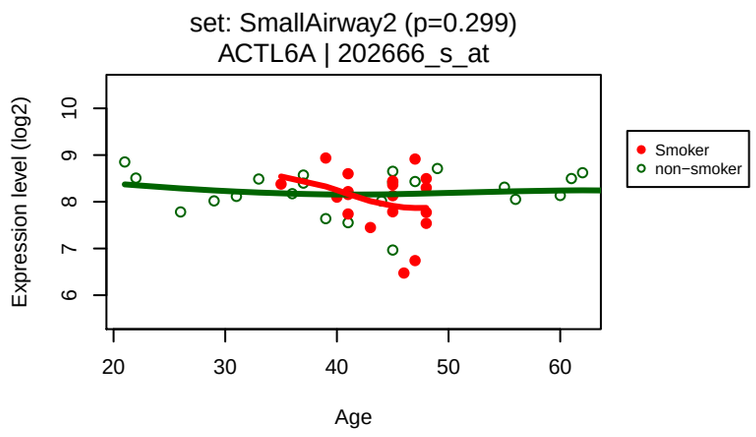
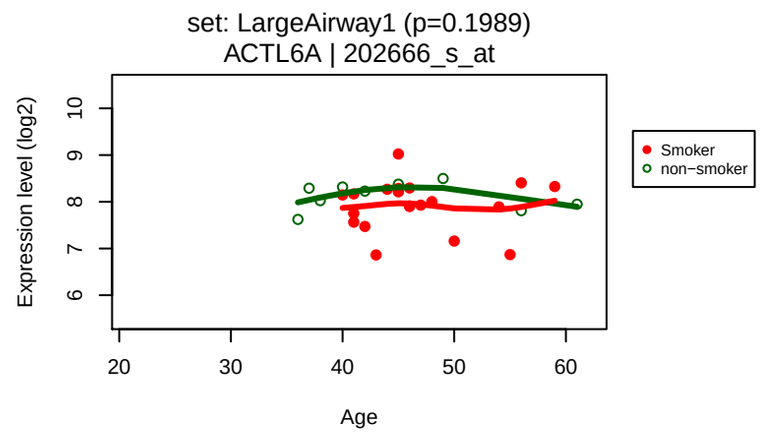

Gene expression of smokers and nonsmokers
gene: ACTL6A | 202666_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0112
p=0.299
p=0.0266
p=0.1989
p=0.972
p=0.0036
p=0.3366
p=0.4478
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

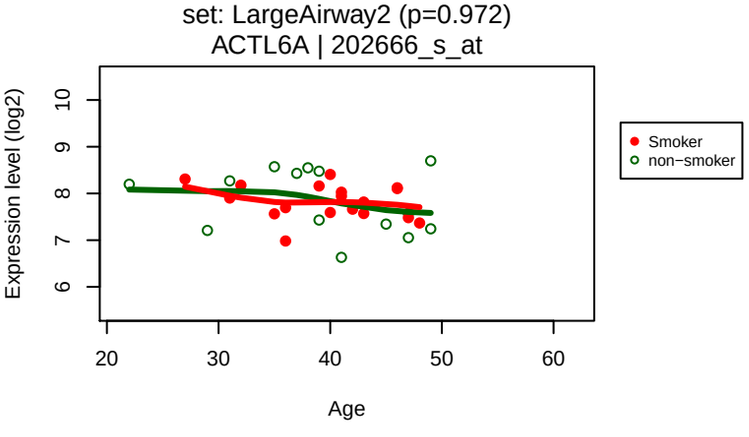

Gene expression ratio (smokers/nonsmokers)
gene: ACTL6A | 202666_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0112
p=0.299
p=0.0266
p=0.1989
p=0.972
p=0.0036
p=0.3366
p=0.4478
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

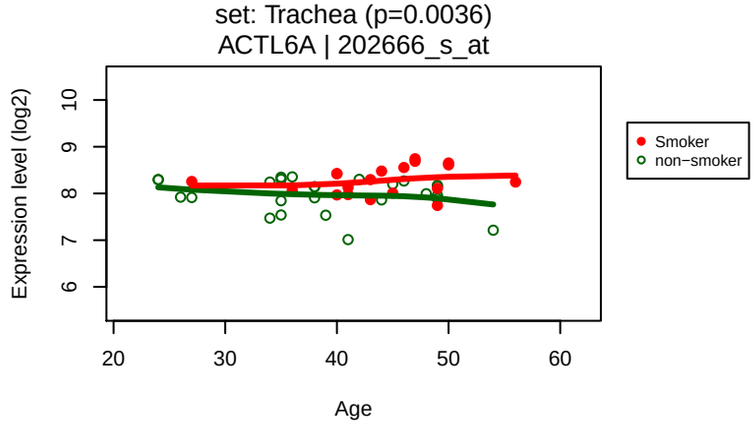

Gene expression ratio in Males
gene: ACTL6A | 202666_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6246
p=0.0948
p=0.023
p=0.121
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTL6A | 202666_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0689
p=0.0164
p=0.1735
p=0.3349
SmallAirway2
LargeAirway2
Trachea
Macrophage2

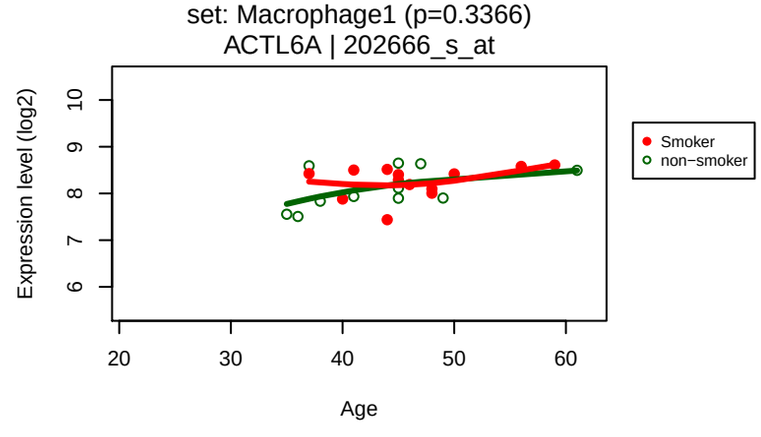

set: SmallAirway1 (p=0.0112)
ACTL6A | 202666_s_at
Expression level (log2)
Age
Smoker
non−smoker

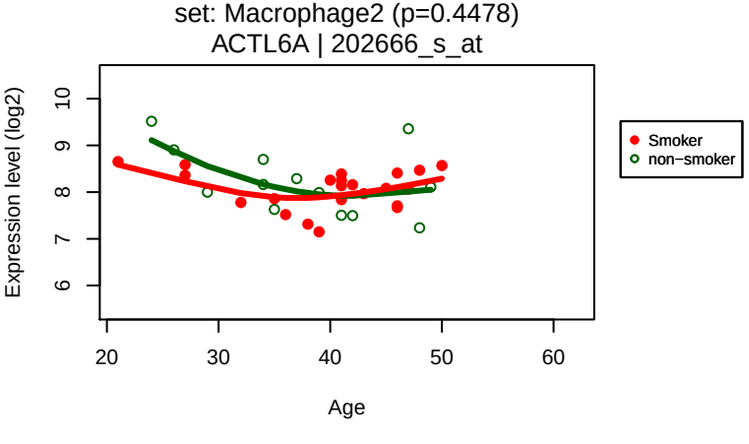

set: SmallAirway2 (p=0.299)
ACTL6A | 202666_s_at
Expression level (log2)
Age
Smoker
non−smoker

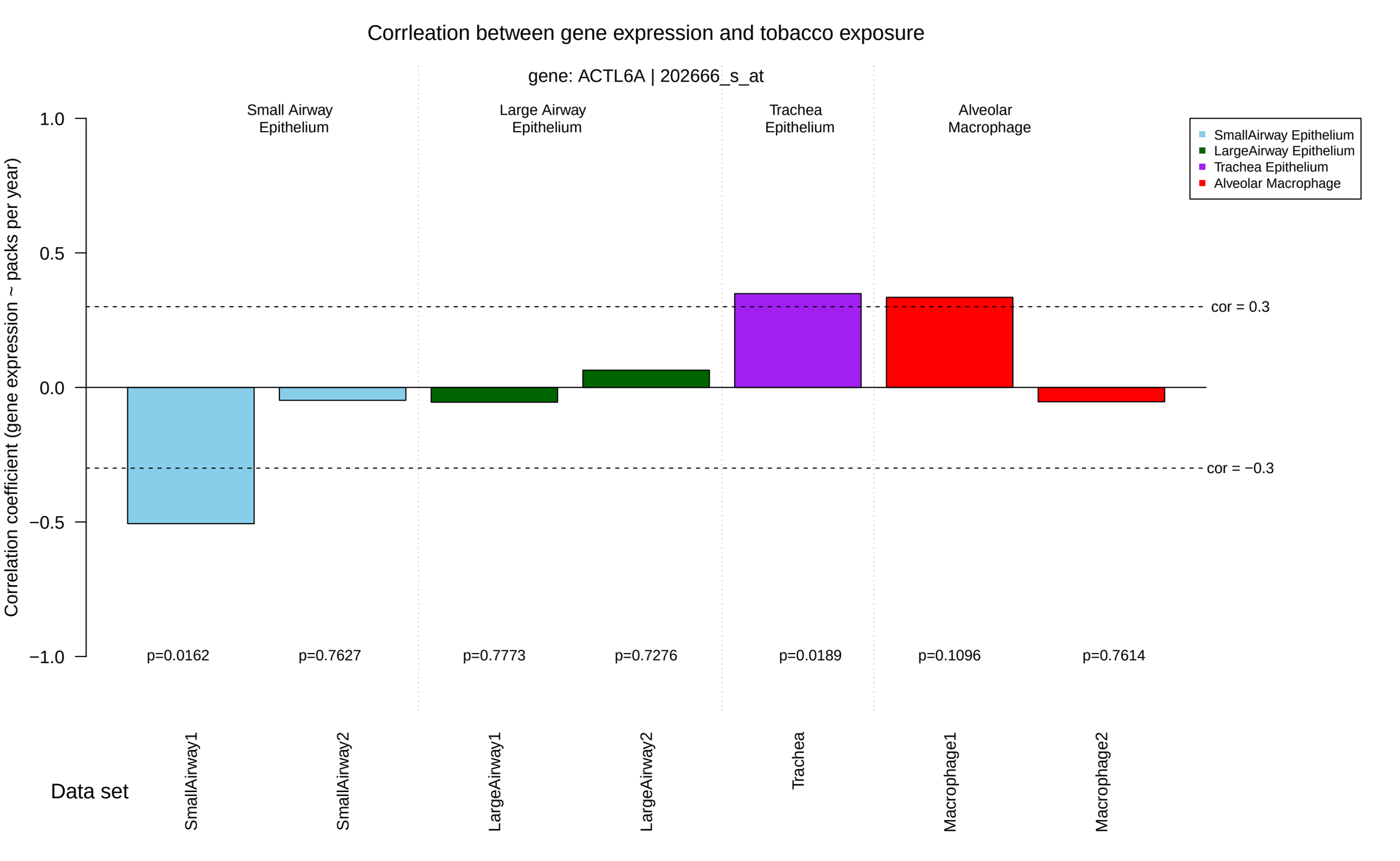

set: LargeAirway1 (p=0.1989)
ACTL6A | 202666_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.972)
ACTL6A | 202666_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0036)
ACTL6A | 202666_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3366)
ACTL6A | 202666_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4478)
ACTL6A | 202666_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTL6A | 202666_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0162
p=0.7627
p=0.7773
p=0.7276
p=0.0189
p=0.1096
p=0.7614
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2